

Pricing NHL Trades on One Scale

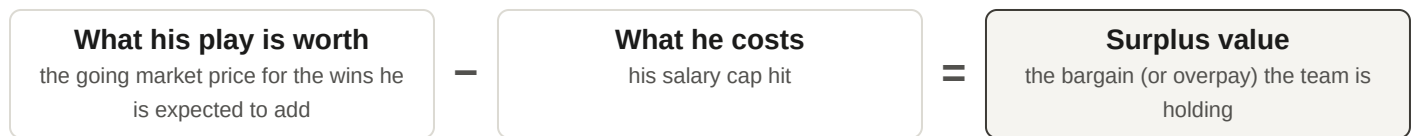
A plain-language overview of the project · Thomas Mihaljevic · September 2026

RESEARCH QUESTION

When NHL teams trade players, draft picks, and prospects, do they price each kind of asset consistently, or do some kinds reliably sell for more or less than they turn out to be worth?

1. The Idea in One Paragraph

A hockey trade is a swap of very different things: a veteran on a signed contract, a draft pick that might become anyone, or a teenager playing junior hockey. Comparing them is like comparing a car, a lottery ticket, and an apprentice, so the project converts each one into the same currency, which is the *surplus value* it gives its team in dollars. Surplus is what the player's hockey would cost to buy on the open market, minus what the team actually pays him under the salary cap. Once each asset is in the same currency, the two sides of any trade can be added up and compared, and then checked against what actually happened on the ice.



For example, a player whose play would cost \$6 million a season on the open market but who is paid \$4 million gives his team \$2 million of surplus that season. The model adds this up over each remaining season, allowing for the chance he declines or leaves the league.

2. The Three Kinds of Assets

ASSET 1

Rostered players

Like buying a used car with a known price tag.

An NHL player on a signed contract. His cost is known (his cap hit); what is uncertain is how well he will play. If he cannot yet sign anywhere when the contract ends, the team keeps his rights for a few more years.

ASSET 2

Draft picks

Like a lottery ticket; higher picks have better odds.

The right to choose a player at a given spot in a future draft. Most late picks never play an NHL game, but the few that do can be very valuable, because drafted players start on cheap entry-level contracts.

ASSET 3

Prospects

Like an apprentice who may or may not make the grade.

A player already drafted but not yet in the NHL, playing in junior, college, the minors, or Europe. Each season he plays tells us more about whether he will make it and how good he will be.

3. What the Project Looks For in the Data

1. **Value each trade on the day it was made.** For each trade from 2018 to 2026, the model prices what each team gave up and what it received, using only information available on that date.
2. **Compare the two sides.** Each trade is expressed as a ratio of value sent to value received. A fair trade sits near 1.
3. **Follow what actually happened.** The prediction is then compared with how the players performed afterward.
4. **Look for patterns.** The question is whether mistakes are random or systematic, for example, whether picks are routinely overpaid for, or whether long contracts, older players, or contending teams buying at the deadline consistently come out behind.

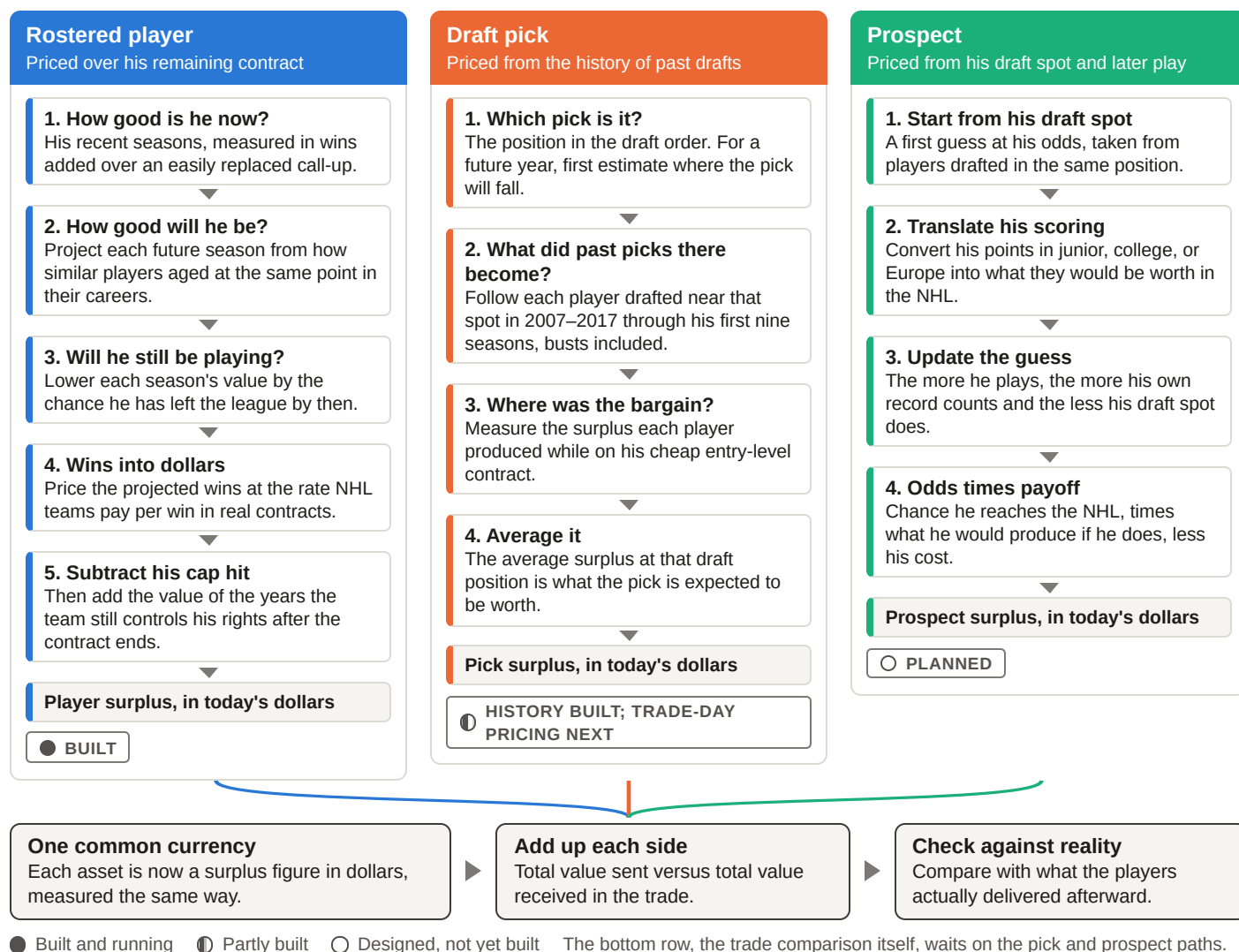
Keeping the test honest. Three rules protect the results:

- **No peeking.** A valuation uses only what was known before the trade date, so the model cannot look smarter than a general manager could have been.
- **A separate ruler.** Outcomes are also checked with a measure built independently from NHL game data (each shot, shift, and penalty), so the model is not grading itself with its own yardstick.
- **Count the misses.** Averages include each pick and prospect who never made it, not just the successes, so the odds are not overstated.

If no pattern appears, that is also a finding: it would suggest teams price these assets sensibly on average.

4. How the Model Values Each Asset

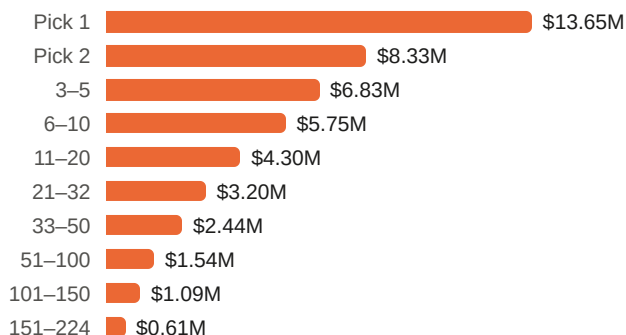
Each kind of asset follows its own path to a dollar figure. All three paths end in the same currency, so they can be combined into a single value for each side of a trade.



5. A Picture of One Result So Far: What a Draft Pick Is Worth

Average surplus value of a draft pick, by draft position

2007–2017 drafts, first nine NHL seasons, expressed at the 2025–26 salary cap (\$95.5M)



How to read it. The first overall pick has historically produced about \$13.7 million of surplus over its first nine seasons, roughly 22 times a late-round pick. The values fall quickly through the first round and then flatten out.

Averages hide the lottery. Only about one in four picks from 151 to 224 plays even a single NHL game in that window. The late-round average is small but positive because a handful of those picks become real players.

These figures are total surplus over the window, not a yearly salary. The same method will price picks in actual trades once the remaining steps are finished.

Data: 980 NHL trades (2018–2026), NHL contracts and salary cap records, published wins-above-replacement estimates, and play-by-play records for 11,870 NHL games.